

CHAPTER 2: SWITCHING CONCEPTS AND VLANS

SWITCH OPERATION

A switch is essentially a high-speed mapping machine. When you power on a switch, the **MAC Address Table** (also known as the Content Addressable Memory or **CAM table**) is empty.

1. The Learning Phase (Source MAC)

- When a device is first plugged in and sends data, the switch examines the **Ethernet Frame**. It ignores the destination for a split second and looks strictly at the **Source MAC Address**.
- The switch records the Source MAC and the physical port it arrived on into its **CAM (Content Addressable Memory) Table**.
- If the MAC address already exists in the table but on a different port, the switch updates the table with the new port (this is how it handles someone moving their laptop to a different desk).

2. The Decision Phase (Destination MAC)

- Once the switch knows where the frame came from, it looks at the Destination MAC Address to decide where the frame should go. It checks its CAM table for a match:
- **Known Unicast:** If the Destination MAC is in the table, the switch knows exactly which port to use. It creates a temporary "private circuit" between the sender and receiver.
- **Unknown Unicast:** If the Destination MAC is not in the table, the switch doesn't just drop the frame. It performs Flooding. It sends the frame out of every single port except the one it came from.
- **Broadcast/Multicast:** If the destination is *FF:FF:FF:FF:FF:FF* (Broadcast), the switch automatically floods it to everyone in that VLAN.

3. The Forwarding Methods

- Modern switches use different internal methods to move the data from the input port to the output port. You can configure these based on whether you want speed or accuracy:

METHOD	HOW IT WORKS	PROS/CONS
Store-and-Forward	The switch waits to receive the <i>entire</i> frame and checks it for errors (CRC) before sending it.	Pro: Error-free. Con: Highest latency.
Cut-Through	The switch reads only the first 6 bytes (Destination MAC) and starts forwarding immediately.	Pro: Lowest latency. Con: Forwards bad/corrupt frames.
Fragment-Free	The switch reads the first 64 bytes (where most collisions occur) before forwarding.	The Middle Ground: Faster than Store-and-Forward, more reliable than Cut-Through.

Mac Address Table

The **MAC Address Table** (also known as the **CAM Table—Content Addressable Memory**) is the "brain" of a Layer 2 switch. While a router uses an IP routing table to find networks, a switch uses this table to find specific physical devices.

A standard MAC address table contains three primary pieces of information for every entry:

- **The MAC Address:** The unique 48-bit hardware identifier of the device.

- **The Port:** The physical interface on the switch (e.g., Fa0/1, Gi1/0/2) where that device is plugged in.
- **The VLAN:** The logical network the device belongs to.

VIRTUAL LOCAL AREA NETWORKS (VLANs)

A VLAN is a logical grouping of devices. Even if 10 people are plugged into the same physical switch, we can "trap" them in separate virtual rooms so they can't hear or talk to each other.

Why do we use them?

1. **Security:** You don't want a guest in the lobby to be able to "see" the server where the company's payroll is stored.
2. **Performance:** In a large network, "broadcasts" (like ARP requests) act like background noise. VLANs keep that noise contained to small groups.
3. **Organization:** You can group people by department (Accounting, Engineering, Sales) rather than by which floor of the building they sit on.

VLAN TRUNKING

If you have two switches and both have users in "VLAN 10 (Accounting)," you need a way to send that traffic between the switches. You *could* run a separate cable for every VLAN, but that's a waste of ports. Instead, we use a **Trunk**.

- **The Trunk Port:** A special port configured to carry traffic for **all** VLANs.
- **802.1Q Tagging:** To keep the data from getting mixed up, the switch adds a small "tag" to the frame as it enters the trunk. This tag says, "Hey, I belong to VLAN 10." When the frame reaches the other switch, the tag is removed, and the frame is delivered to the correct VLAN.

INTER-VLAN ROUTING

Devices in different VLANs cannot talk to each other by default. They are logically on different networks. To get them to communicate, you need a "Layer 3" device (a Router or a Multilayer Switch).

The Three Methods:

1. **Legacy:** One physical cable per VLAN between a switch and a router. (Nobody does this anymore; it's a waste of cables).
2. **Router-on-a-Stick (ROAS):** A single physical trunk cable connects the switch to the router. The router uses "sub-interfaces" (like Gig0/0.10 and Gig0/0.20) to act as the gateway for each VLAN.
3. **Layer 3 Switching (SVI):** This is the modern way. The switch itself handles the routing internally using **Switch Virtual Interfaces**. It's much faster because the data never has to leave the switch.

References:

- Forouzan, B. A. (2017). *Data Communications and Networking* (5th ed.). McGraw-Hill Education.
- Kurose, J. F., & Ross, K. W. (2021). *Computer Networking: A Top-Down Approach* (8th ed.). Pearson Education.